

Minimal Demo of Bayesian Measurement Error Correction

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Express the joint model for observables and latents through three component models

```
jags.code <- "  
model {  
  for (i in 1:n) {  
    y[i] ~ dbern(p[i])  ### outcome model  
    logit(p[i]) <- beta0 + beta1 * x[i]  
  }  
  for (i in 1:n) {  
    x.str[i] ~ dnorm(x[i], prc.ME) ### meas err model  
  }  
  for (i in 1:n) {  
    x[i] ~ dnorm(mu, prc.X)  ### exposure model  
  }  
  
  sd.ME ~ dunif(a,b)  # prior (expert opinion) on ME magnitude  
  prc.ME <- pow(sd.ME, -2)  
  
  # diffuse priors on other parameters  
  mu ~ dnorm(0, 0.0001)  
  prc.X ~ dgamma(0.001, 0.001)  
  beta0 ~ dnorm(0, 0.0001)  
  beta1 ~ dnorm(0, 0.0001)  
}  
"
```

We design the test world and data

```
require(rje)  
  
## Loading required package: rje  
  
### the truth in our made-up world  
set.seed(13)  
x <- rnorm(400)  
x.str <- rnorm(400, mean=x, sd=0.8)  
y <- rbinom(400, size=1, prob=expit(-1+x))
```

Ideal and naive analyses:

```
summary(glm(y~x, family=binomial))
```

```
##
## Call:
## glm(formula = y ~ x, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.2316     0.1402  -8.785  < 2e-16 ***
## x              1.2336     0.1544   7.989 1.36e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 474.36  on 399  degrees of freedom
## Residual deviance: 381.57  on 398  degrees of freedom
## AIC: 385.57
##
## Number of Fisher Scoring iterations: 5
```

```
summary(glm(y~x.str, family=binomial))
```

```
##
## Call:
## glm(formula = y ~ x.str, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.1102     0.1279  -8.681  < 2e-16 ***
## x.str          0.7358     0.1106   6.653 2.87e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 474.36  on 399  degrees of freedom
## Residual deviance: 418.76  on 398  degrees of freedom
## AIC: 422.76
##
## Number of Fisher Scoring iterations: 4
```

Now fit the Bayesian model with expert opinion about the ME magnitude:

```
require(rjags)
```

```
## Loading required package: rjags
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.1
```

```
## Loaded modules: basemod,bugs
```

```
jags.mod <- jags.model(textConnection(jags.code),  
  data=list(n=length(y), y=y, x.str=x.str, a=0.6, b=0.9),  
  n.chains=3)
```

```
## Compiling model graph  
##   Resolving undeclared variables  
##   Allocating nodes  
## Graph information:  
##   Observed stochastic nodes: 800  
##   Unobserved stochastic nodes: 405  
##   Total graph size: 2414  
##  
## Initializing model
```

```
set.seed(17)  
update(jags.mod, 5000) ## burn-in MCMC iterations  
jags.opt <- coda.samples(jags.mod, n.iter=400000, thin=20,  
  variable.names=c("beta0", "beta1", "sd.ME"))
```

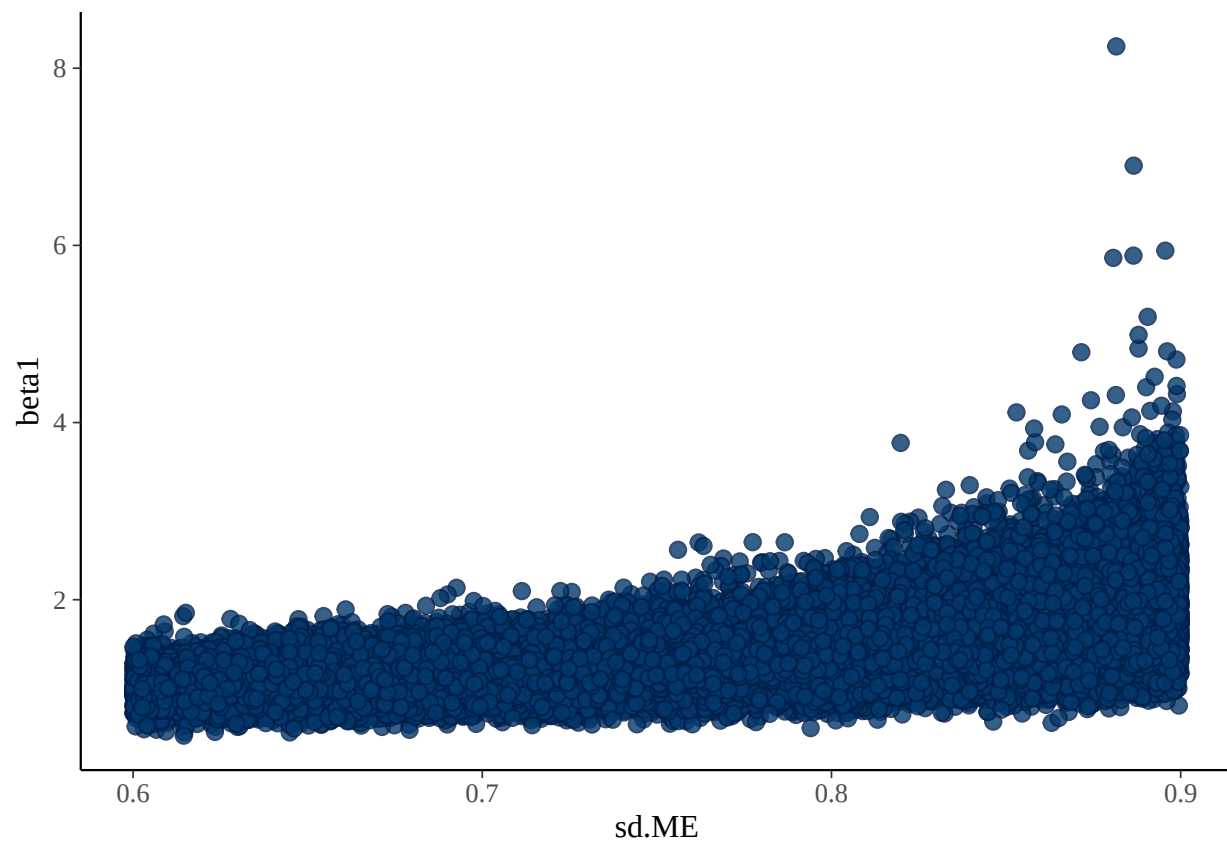
Parameter inference:

```
summary(jags.opt)
```

```
##  
## Iterations = 6020:406000  
## Thinning interval = 20  
## Number of chains = 3  
## Sample size per chain = 20000  
##  
## 1. Empirical mean and standard deviation for each variable,  
##    plus standard error of the mean:  
##  
##           Mean      SD Naive SE Time-series SE  
## beta0 -1.2861 0.19510 0.0007965      0.001018  
## beta1  1.4337 0.41615 0.0016989      0.002824  
## sd.ME  0.7822 0.08521 0.0003479      0.000645  
##  
## 2. Quantiles for each variable:  
##  
##           2.5%      25%      50%      75%      97.5%  
## beta0 -1.7192 -1.3994 -1.2690 -1.1532 -0.9547  
## beta1  0.8325  1.1369  1.3653  1.6561  2.4168  
## sd.ME  0.6136  0.7159  0.7983  0.8571  0.8962
```

Joint posterior on measurement error SD and parameter of interest:

```
require(bayesplot)  
#pdf("minimal-Bayes-fig.pdf")  
mcmc_scatter(jags.opt, "sd.ME", "beta1")
```



```
#graphics.off()
```